

Agentic Coding Overview

What to use, how it works, and how to choose

Cédric Jung Jozef Janus

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1 What can I use?

AQUEDUCT at TU Wien:
the models it serves, the
hardware behind them,
and how you get a key.

2 How does it work?

A coding agent is a
model inside a harness.
We look at both, and at
why they are trained
together.

3 How do I choose?

Where inference runs,
what it costs, and what
six benchmarks do and
do not tell you.

One idea runs through all three.

A coding agent is a whole system. Buy it and measure it as one.

What AQUEDUCT serves today

Four LLMs on the public overview, checked on 30 September 2026

Model	Context	Runs on	Good to know
DeepSeek V4 Flash 284B	393k	1 replica on H200, not quantized	Text only. Moving to RTX PRO 6000 Blackwell
Qwen 3.6 35B	262k	4 replicas, one A40 each, AWQ	Small, fast, accepts images
Qwen 3.5 397B	200k	2 replicas, 4 RTX PRO 6000 each, GPTQ-Int4	Retires today
GLM 5.2 744B Preview	262k	1 replica on 8 MI300X, FP8	Preview, the largest model on offer

In transition

DeepSeek V4 Flash takes over the Blackwell GPUs from Qwen 3.5, which frees the H200 node.

What may come next

Named in the 17 September release note. Nothing on this slide is deployed yet

Candidate	Hardware discussed	What it would add
Qwen3.8-27B	The freed H200 node, possibly four instances	A newer small model, in several copies
DeepSeek V4 Flash Vision Exp GLM-5.3-Flash	RTX PRO 6000 Blackwell RTX PRO 6000 Blackwell	Image input for DeepSeek The best open agentic scores we show
DeepSeek V4.1 Flash or GLM-5.3	AMD MI300X	A successor on the largest node

The one rule

A model exists for you once `GET /v1/models` returns it. Release notes describe intentions.

Big models need big memory

The hardware decides which models can be served, and how many copies of each

Accelerator	Per GPU	Per replica	What runs there
NVIDIA A40	48 GB	1 GPU = 48 GB	Qwen 3.6 35B at 4 bit, four copies
RTX PRO 6000 Blackwell	96 GB	4 GPUs = 384 GB	Qwen 3.5 397B at 4 bit until today, then DeepSeek V4 Flash
NVIDIA H200	141 GB	count not published	DeepSeek V4 Flash 284B, not quantized
AMD MI300X	192 GB	8 GPUs $\approx$ 1.5 TB	GLM 5.2 744B at 8 bit

Rule of thumb

Weights take about 1 GB per billion parameters at 8 bit and half of that at 4 bit, before any room for context. A 744B model at 8 bit is roughly 744 GB, which is why it sits alone on the 1.5 TB node.

Sources: TU Wien: [AQUEDUCT overview](#), [release notes](#). The rule of thumb is our own arithmetic.

How you get access

Single sign-on and a personal API key. Students have to be allowlisted first

1. **Researchers, lecturers and staff** sign in with TU Wien SSO at aqueduct.ai.datalab.tuwien.ac.at and create a personal API key.
2. **Students** are not in that group by default. Ask to be allowlisted first, through your supervisor or the dataLAB team on [Matrix](#).
3. **Point your harness at it.** Any OpenAI-compatible client works, and Pi, OpenCode, Crush and Zed have setup guides.
4. **Ask the API what is there.** The model list it returns is the truth.

```
export BASE=https://aqueduct.ai.datalab.tuwien.ac.at/v1
curl -H "Authorization: Bearer $KEY" $BASE/models
```

A coding agent is a model plus a harness

The model decides what to do next, and the harness is the part that actually does it

Model: decides

- Reads the task and the code
- Chooses the next edit or tool call
- Makes sense of what comes back

GPT, Claude, Qwen, DeepSeek, GLM

Harness: acts

- Gathers context, offers tools
- Edits files and runs the tests
- Keeps state and permissions

Codex, Claude Code, OpenCode, Mistral Vibe

You ask → harness builds context → model picks an action → harness runs it → repeat

The model is the engine. The harness is the car built around it.

Source: [OpenAI](#), *Unrolling the Codex agent loop*

You can mix and match, up to a point

The harness is a separate choice, and several of them already talk to **AQUEDUCT**

Harness	Made by	Models it drives
Codex	OpenAI	GPT models, and GPT-5-Codex was tuned for it
Claude Code	Anthropic	Claude models, plus skills, subagents and MCP
OpenCode	Open source	75+ providers, with an AQUEDUCT setup guide
Mistral Vibe	Mistral, open source	Mistral by default, other providers configurable
Pi, Crush, Zed	Open source	Each has an AQUEDUCT setup guide

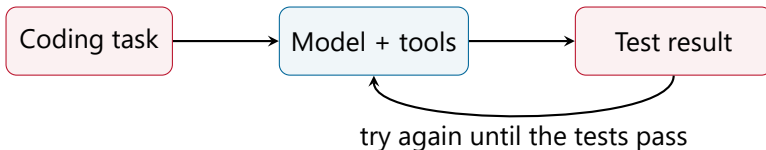
Compatible is not the same as well matched

Any OpenAI-compatible harness can call an AQUEDUCT model. How well it works depends on the model's tool calling, its context window, and the harness it was trained around.

Sources: [OpenCode](#), [Mistral Vibe](#), [Claude Code foundations](#), [AQUEDUCT integrations](#)

The model has learned its harness

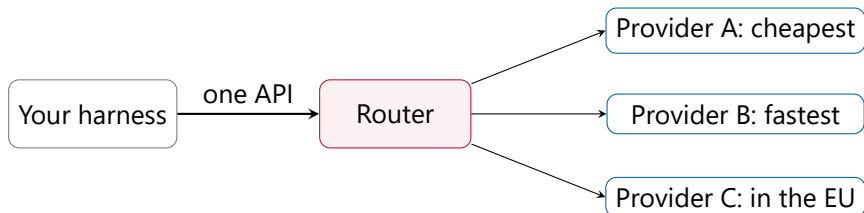
Post-training runs inside the agent loop, and Codex is the clearest example



- Coding models no longer learn only from prompt and answer. They learn from whole trajectories: inspect, edit, run the tests, read the failure, try again.
- OpenAI says this openly. codex-1 and GPT-5-Codex were trained with reinforcement learning on real coding tasks, and GPT-5-Codex is tuned for Codex itself.
- You can see it in the scores. On Terminal-Bench 4.0, Grok 4.7 gets about 26% in a neutral harness and 38% in xAI's own. Opus 5.5 gets 59.6% and 66.4%.

Why routers exist

One API in front of many models and many providers



- The **model** is which network answers. The **provider** is who runs it, and the same open-weight model is often sold by several at different prices, speeds and regions.
- A router gives you one integration and one bill. It picks a provider for each request and switches to another when one is down.
- So the full stack has five layers: harness, router, model, provider, region.

What you get

- One API and key for all models
- Automatic provider fallback
- Provider allowlists and blocklists
- Bring your own provider key

How you steer it

- Sort by price or throughput
- `:floor` is cheapest, `:nitro` fastest
- Pin a specific provider order
- EU-only endpoint for enterprise

Routing is global by default. **Before you send unpublished code, check which provider and region will serve it.**

Sources: [OpenRouter quickstart](#), [nitro and floor shortcuts](#), [EU in-region routing](#)

Three services, three different kinds of control

Service	What it is	What it guarantees
EUrouter	Router across providers	EU processing for every request, plus filters for region, EU ownership and retention
Eden AI	Router across providers	A separate EU endpoint. Models without EU hosting are hidden, never rerouted
Scaleway	One provider, many models	Serverless models hosted in Paris. It can itself sit behind a router

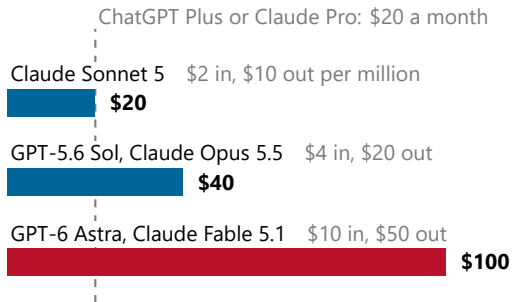
EU-hosted is not EU-owned

A US company can run a data centre in Frankfurt. Decide three things separately: where the GPU is, who owns the provider, and what they keep of your prompts.

Sources: [EUrouter docs](#), [Eden AI EU](#), [Scaleway FAQ](#)

A subscription goes further than the API

List prices before caching: 5M input and 1M output tokens against a \$20 plan



- An agent sends code, history and test logs again on every step. Millions of tokens go quickly.
- A plan includes Codex or Claude Code, within five-hour and weekly limits.
- Same harness, two bills. Claude Code on a Pro login uses the plan. On an API key it is metered.

Working interactively? Start with the subscription. Pay per token when you need automation, your own harness, or control over where inference runs.

Sources: [Codex with a ChatGPT plan](#), [Claude Code with Pro](#), [OpenAI pricing](#), [Anthropic pricing](#)

Which benchmark answers which question?

One chart per benchmark follows

Your question	Benchmark	Metric
Can it write a professional document?	GDPval-AA v2.1	Elo
Can it finish a long engineering task?	DeepSWE v1.1	% resolved
Can it get things done in a terminal?	Terminal-Bench 4.0 and 2.1	% passed
Does it reason like an expert?	Humanity's Last Exam	% correct
Does it know graduate-level science?	GPQA Diamond	% correct

How to read the charts

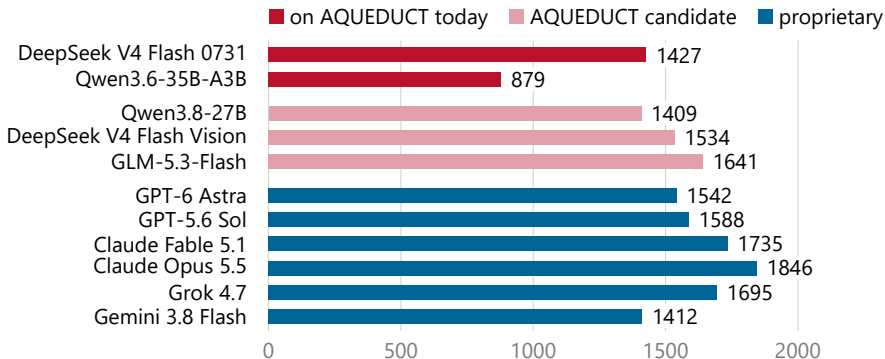
■ on AQUEDUCT today ■ AQUEDUCT candidate ■ proprietary

N/A means nobody has published that exact result. It does not mean zero.

Sources: [Artificial Analysis](#), [DeepSWE](#), [Terminal-Bench](#)

Knowledge work: open models are mid-pack

GDPval-AA v2.1 (Elo, higher is better)

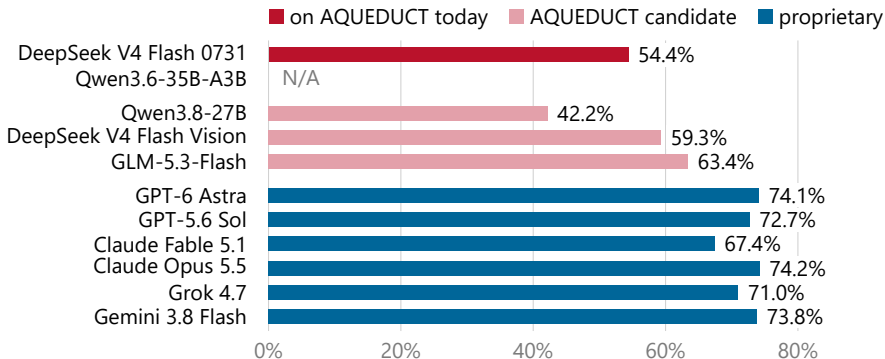


One evaluator and one version for all eleven models, so this is the cleanest comparison we have. GLM-5.3-Flash is ahead of both GPT models. It measures finished professional documents, not academic writing.

Sources: [Artificial Analysis GDPval-AA](#), [Qwen3.8 xhigh result](#)

Long engineering tasks: a 10 to 20 point gap

DeepSWE v1.1 (share of tasks resolved, higher is better)

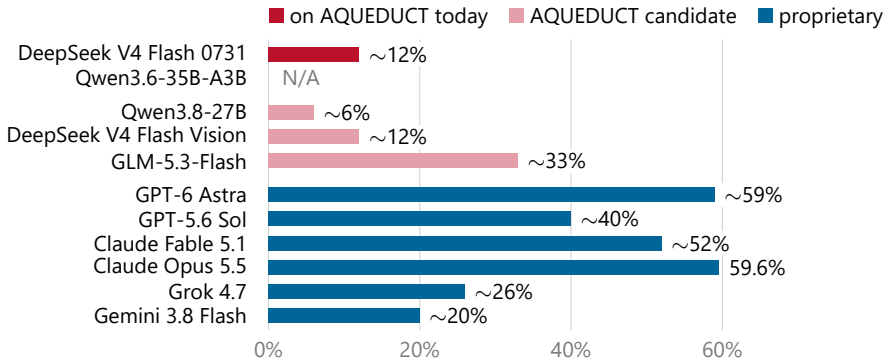


Mixed provenance: vendors report in their own harness, the leaderboard uses a neutral one, and the Fable 5.1 figure is the one OpenAI published. The leaderboard's own error bars are 3 to 4 points, so the top four are a tie.

Sources: [DeepSWE](#), [Qwen3.8 card](#), [DeepSeek Vision](#), [GLM Flash](#), [Gemini](#), [Astra](#) and [Fable 5.1](#)

Hard terminal tasks spread the field

Terminal-Bench 4.0 (pass rate in the Artificial Analysis setup, higher is better)

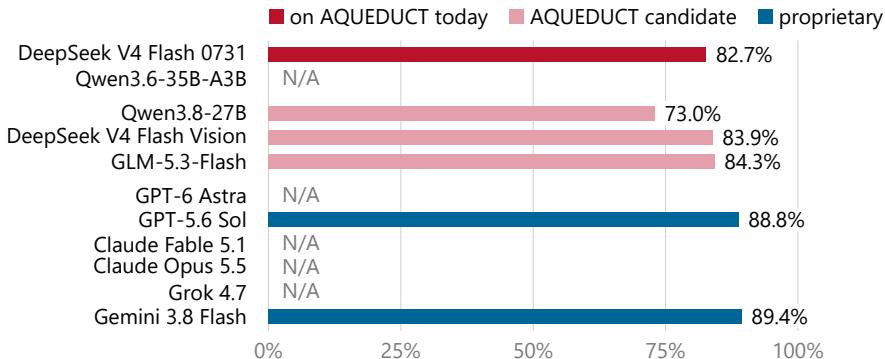


66 tasks, one neutral harness for every model. This is where the harness effect shows: vendors score higher in their own (Opus 5.5 66.4%, Grok 4.7 about 38%). Claude Sonnet 5.5, not charted, leads this board at 63.6%.

Sources: [Artificial Analysis Terminal-Bench 4.0](#), [Astra](#) and [Fable 5.1](#), [Qwen3.8 xhigh](#) result

The older version is almost solved

Terminal-Bench 2.1 (pass rate as published by each vendor, higher is better)

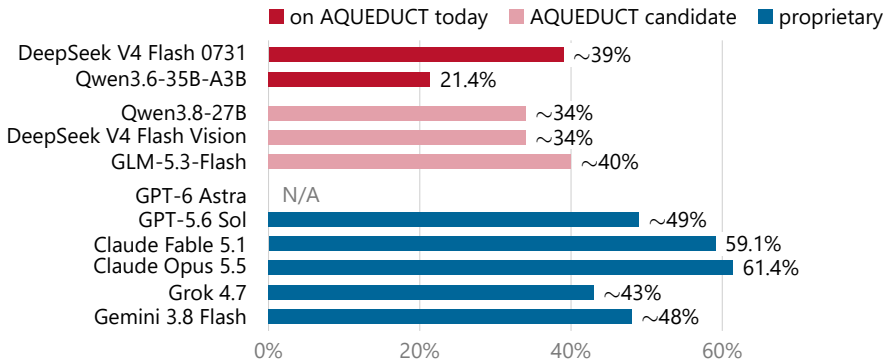


The same DeepSeek V4 Flash that reaches 82.7% here gets about 12% on version 4.0. Never compare across versions. The newest models no longer report 2.1 at all, which is why half of the blue bars are missing.

Sources: [Terminal-Bench](#), [Qwen3.8 card](#), [DeepSeek Vision](#), [GLM Flash](#), [Gemini](#)

Expert reasoning: a 20-point gap remains

Humanity's Last Exam (text-only questions answered correctly, higher is better)

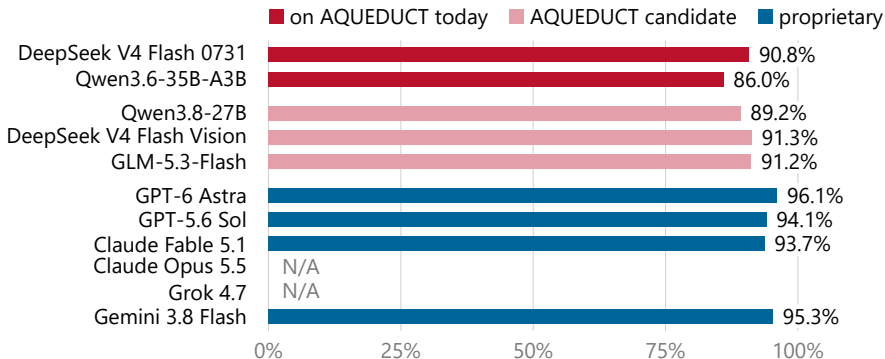


Artificial Analysis, 2,158 text-only questions. Qwen3.6 is from its model card. There is no comparable GPT-6 Astra result yet: OpenAI reports 57.2% with tools, which is a different test. Even the best model misses four in ten.

Sources: [Artificial Analysis HLE](#), [Qwen3.8 xhigh result](#), [Qwen3.6 card](#)

Graduate science: the test is saturated

GPQA Diamond (accuracy on 198 questions, higher is better)



Everything sits between 86% and 96%, so this test no longer separates models, and a 35B model on four A40s is within ten points of the frontier. Evaluators are mixed: Qwen figures are vendor-reported, Fable 5.1 is as published by OpenAI.

Sources: [Artificial Analysis GPQA](#), [Qwen3.8 card](#), [Qwen3.6 card](#), [Fable 5.1](#)

1. **Start with what is already here.** Get an AQUEDUCT key, point OpenCode or Pi at it, and try DeepSeek V4 Flash for coding. Use Qwen 3.6 when you need image input or many calls in parallel.
2. **Judge the pair, never the model alone.** A score belongs to a model, a harness and a reasoning budget together. Compare within one benchmark version and one setup.
3. **Match the bill to the job.** For interactive work on frontier models, a \$20 subscription comes first. For automation or your own harness, use an API, and check provider, region and retention before you send unpublished code.

Everything in this deck is a snapshot of 30 September 2026. It will age quickly.

- TU Wien AQUEDUCT: [overview](#), [release notes](#), [integrations](#), [model compatibility](#)
- OpenAI: [Unrolling the Codex agent loop](#) and [GPT-5-Codex system-card addendum](#)
- Pan et al., SWE-Gym (ICML 2025) and Da et al., Agent-RLVR
- Harnesses: [OpenCode](#), [Mistral Vibe](#), [Claude Code foundations](#)
- OpenRouter [quickstart](#) and [EU in-region routing](#)
- EUrouter [documentation](#), [Eden AI EU](#), [Scaleway FAQ](#)
- Plans: [Codex with a ChatGPT plan](#), [Claude Code with Pro or Max](#)
- API prices: [OpenAI](#), [Anthropic](#)

Built from the research notes in [documentation/](#), including the live check of 30 September 2026.
Links are clickable in the PDF.

Exact versions and configurations matter

- Artificial Analysis: [GDPval-AA v2.1](#), [Terminal-Bench 4.0](#), [Humanity's Last Exam](#), [GPQA Diamond](#), [Benchmarking GPT-6 Astra](#)
- DataCurve: [DeepSWE v1.1](#) and the [Terminal-Bench project](#)
- DataCamp: [GPT-6 Astra vs Claude Fable 5.1](#) (collects the vendor-published figures)
- Model cards: [Qwen3.8-27B](#), [Qwen3.6-35B-A3B](#), [DeepSeek V4 Flash Vision](#), [GLM-5.3-Flash](#), [Gemini 3.8 Flash](#)

Keep in mind

A benchmark on AQUEDUCT's quantized deployments may differ from these published figures, which are for the original weights.